

1. Evaluate $\int \frac{x^3}{x^4 - 2} dx$.

$$u = x^4 - 2$$

$$du = 4x^3 dx$$

$$\Leftrightarrow \frac{1}{4} du = x^3 dx$$

$$= \frac{1}{4} \int \frac{1}{u} du$$

$$= \frac{1}{4} \ln|u| + C$$

$$= \frac{1}{4} \ln|x^4 - 2| + C$$

2. Evaluate $\int x^2 \ln x dx$.

$$u = \ln x$$

$$dv = x^2 dx$$

$$du = \frac{1}{x} dx$$

$$v = \frac{1}{3} x^3$$

$$= \frac{1}{3} x^3 \ln x - \int \frac{1}{x} \cdot \frac{1}{3} x^3 dx$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{3} \int x^2 dx$$

$$= \frac{1}{3} x^3 \ln x - \frac{1}{9} x^3 + C$$